

MAGNETIC FIELD

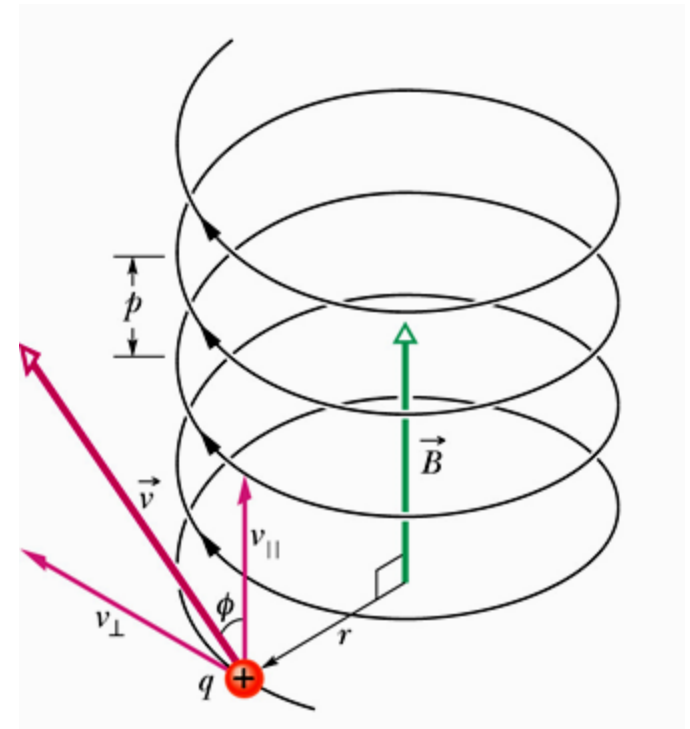
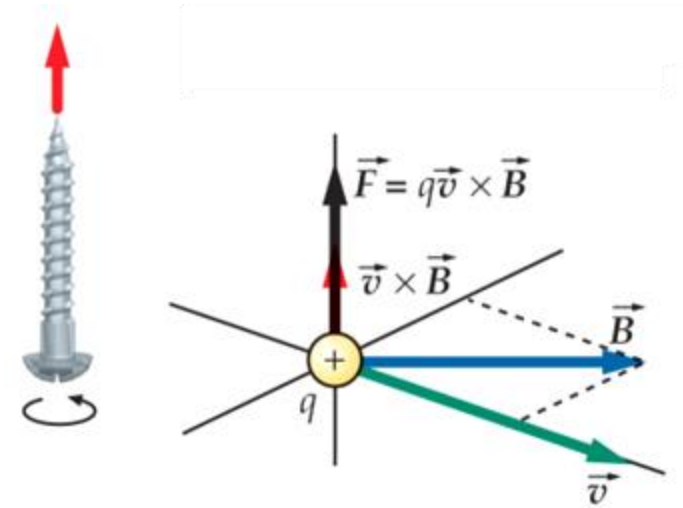
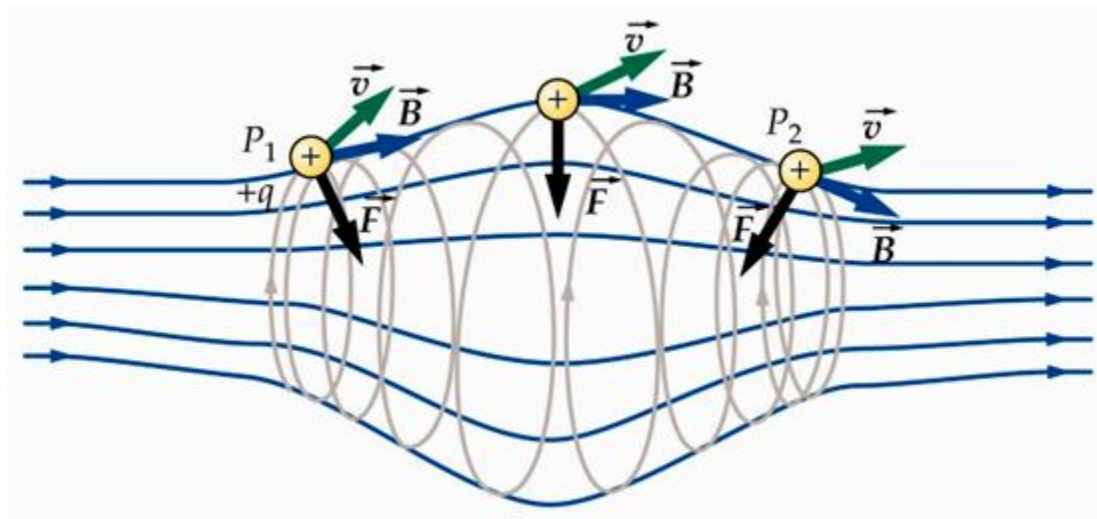
SOURCES OF MAGNETIC FIELD

Definition of the vector of magnetic induction (magnetic B field vector)

$$\vec{F} = q(\vec{v} \times \vec{B})$$

Unit of B is 1T (tesla)
 $1\text{T} = 1\text{N}/\text{Am}$

Magnetic field deflects the trajectory of the electric charge.

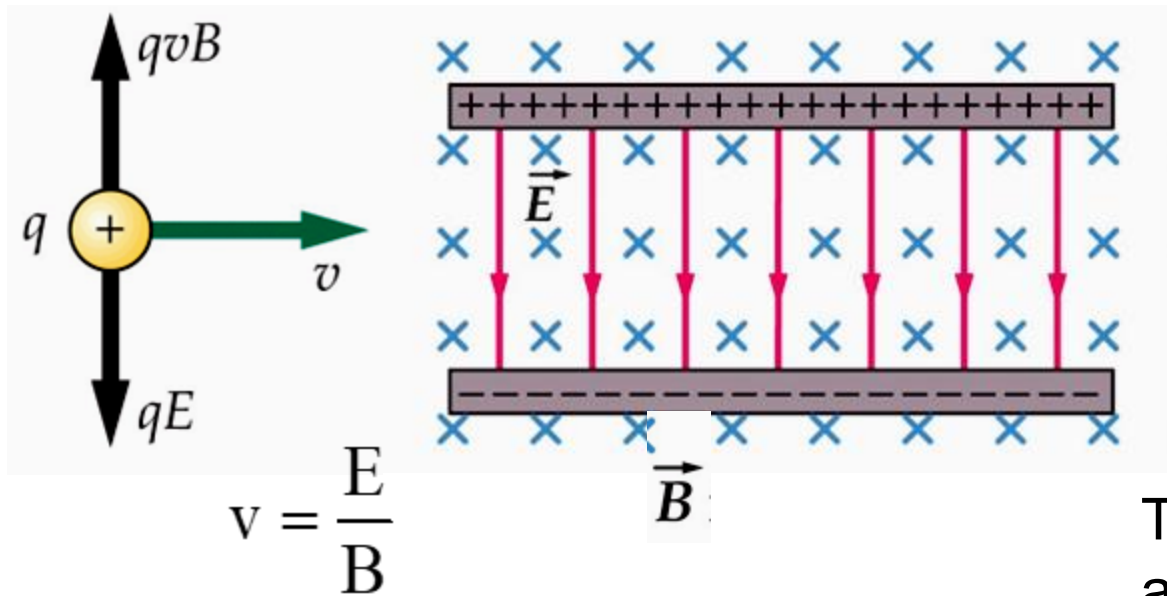


Lorentz force

- Force acting on a charge travelling across the region in which both fields, electric and magnetic exist

$$\vec{F} = q(\vec{v} \times \vec{B} + \vec{E})$$

Movement in crossed fields, i.e., when $\vec{B} \perp \vec{E}$ if



$$\vec{v} \times \vec{B} = -\vec{E}$$

then

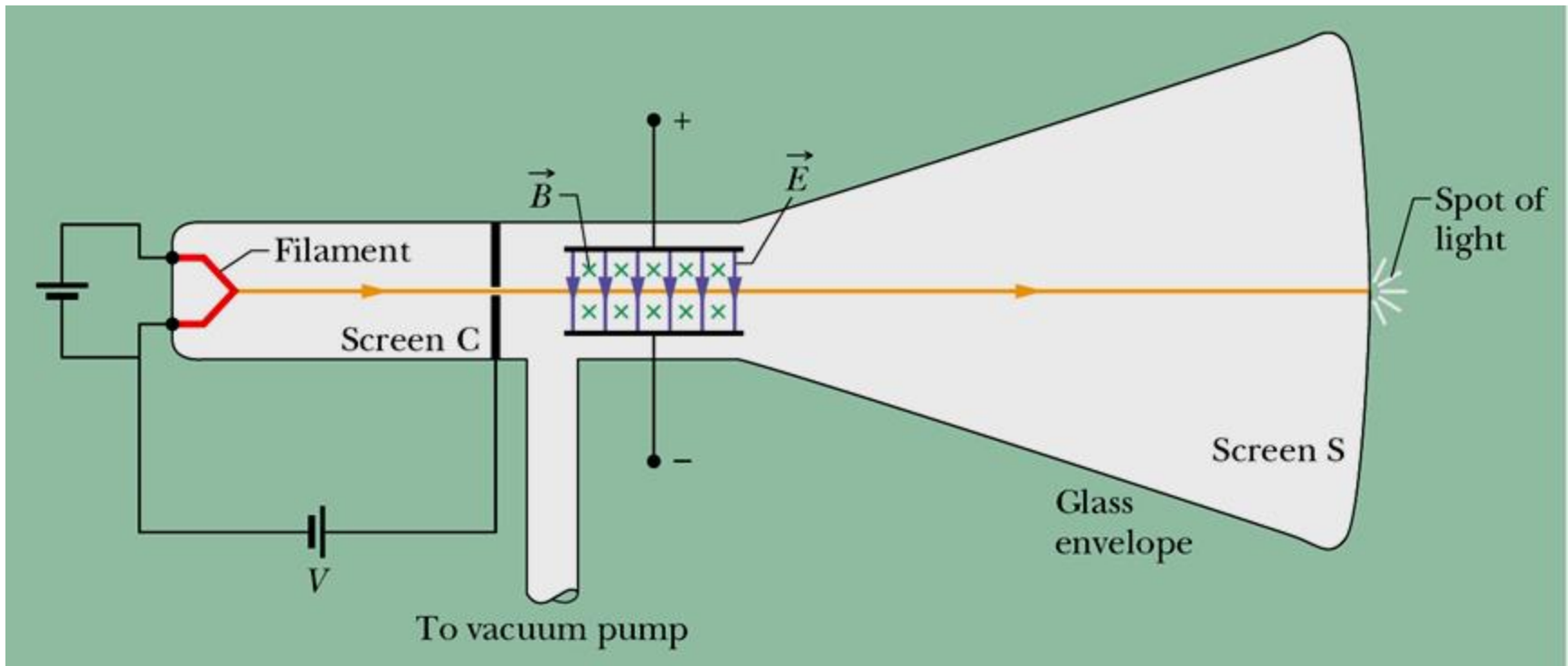
$$\vec{F} = 0$$

Trajectory and velocity of a particle do not change

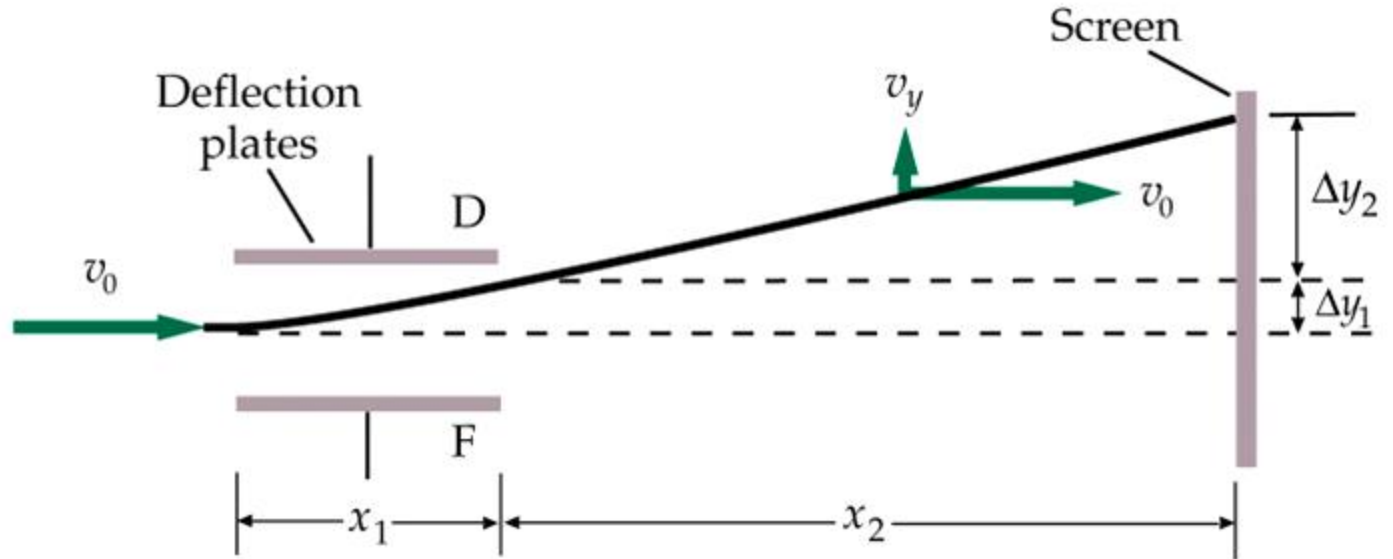
Thomson's experiment

- In 1897 J.J. Thomson, Cambridge, determined q/m for an electron; discovery of the electron

$$\frac{q}{m} = 17.56 \cdot 10^{10} \frac{C}{kg}$$



Homework 10.1

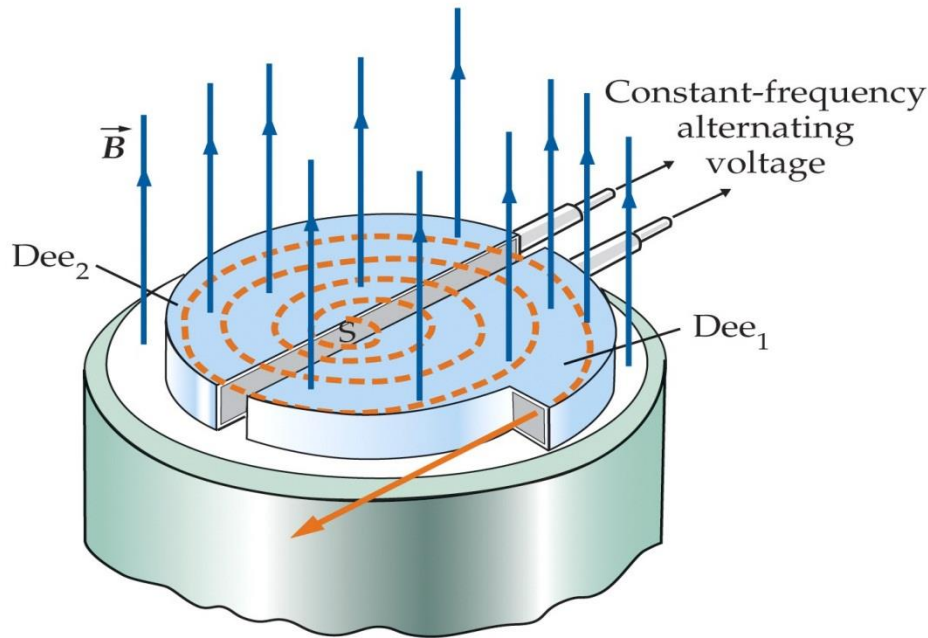


Electron beam passes without deflection through a vacuum space of the oscilloscope when the electric field equals to 3000 V/m, and a magnetic induction is 1.4 Gs; 1Gs (gauss) = 10^{-4} T. The length of deflection plates is 4 cm, while a distance between the end of the plates and the screen equals to 30 cm. Find a net deflection of the electron beam when the magnetic field is turned off.

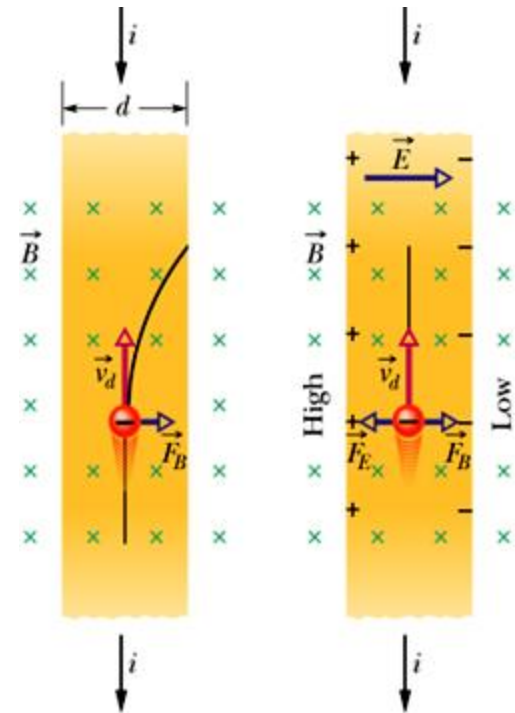
Homework 10.2

Based on textbook HRW, vol.3 or other sources prepare the following two topics:

- cyclotron and synchrotron



- Hall effect



Give examples of their applications

Definition of the electric current

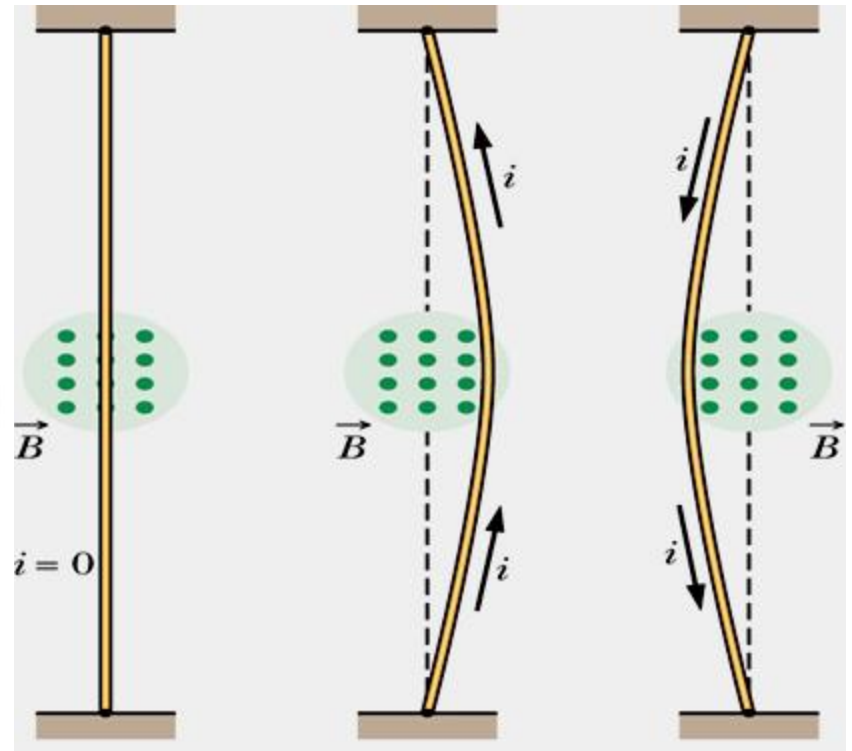
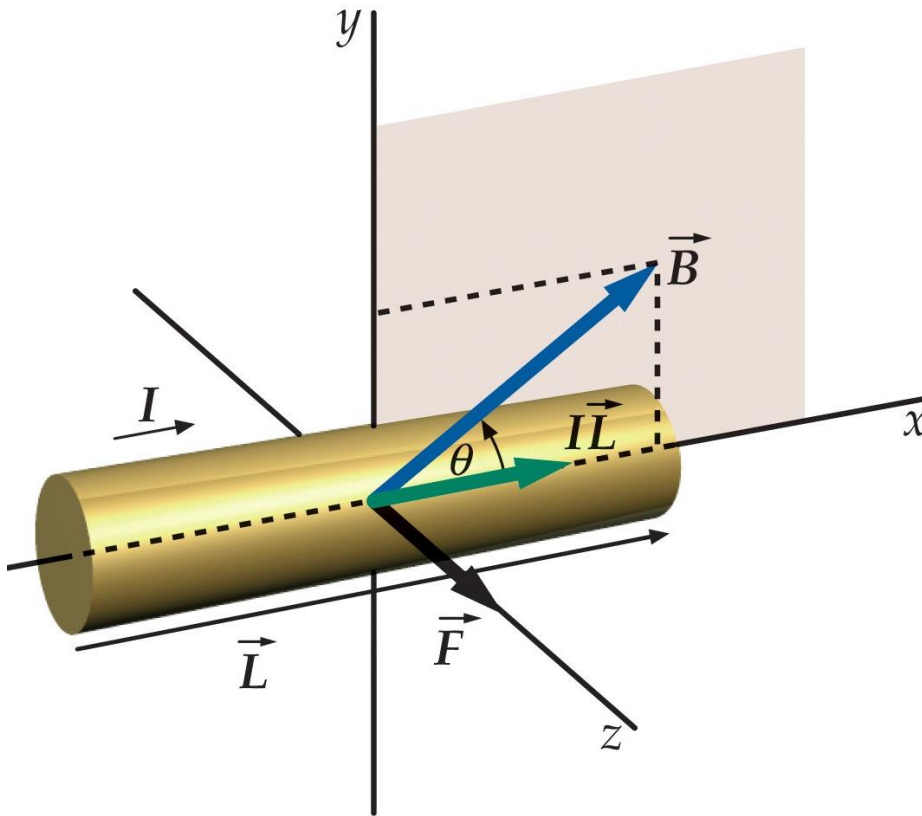
$$i = \frac{dq}{dt}$$

- ❑ Electric current is a scalar quantity
- ❑ The unit of the electric current is $1\text{A}=1\text{C/s}$
(but it is not a definition of 1A!)
- ❑ It is assumed that the electric current is a collective movement of positive charges. In reality, the conduction electrons move in metallic conductors

Electrodynamic force

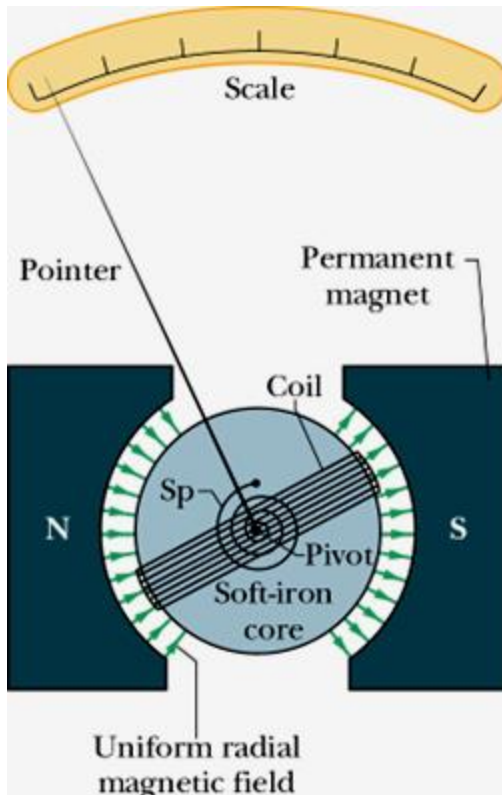
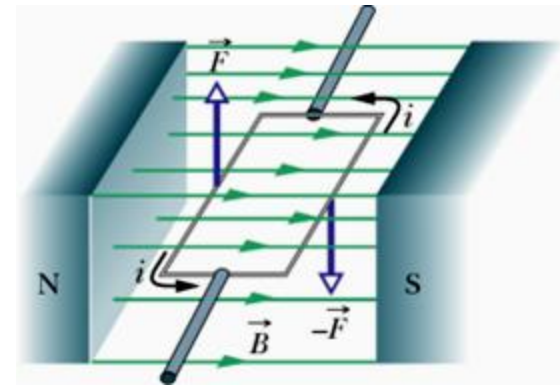
Force acting on a conductor carrying the electric current

$$\vec{F} = i(\vec{L} \times \vec{B})$$



Torque on a current loop

- Electric motor – a rectangular loop carrying a current in the magnetic field



- Analog measuring instruments—
voltmeter, ammeter, galvanometer

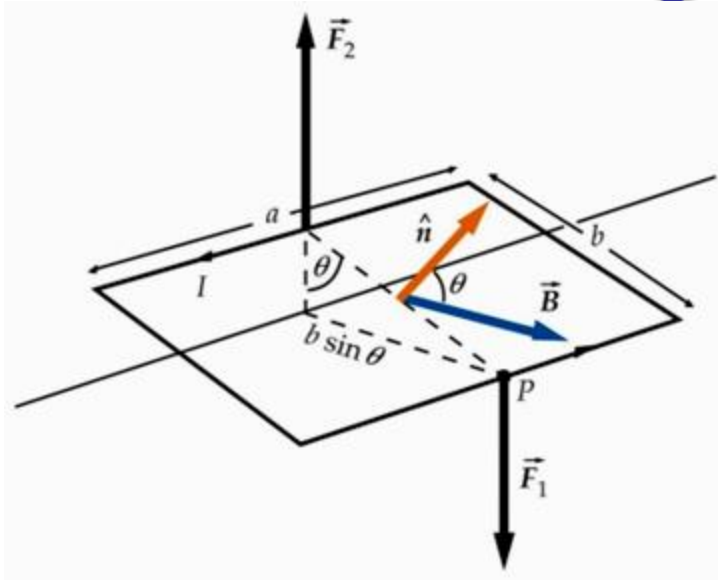
A loop (a frame) with the electric current placed in the external magnetic field will be rotated due to a torque acting on it

$$\vec{\tau} = \vec{\mu} \times \vec{B} \quad \vec{\mu} - \text{magnetic moment}$$

Remainder: for the electric dipole

$$\vec{\tau} = \vec{p} \times \vec{E}$$

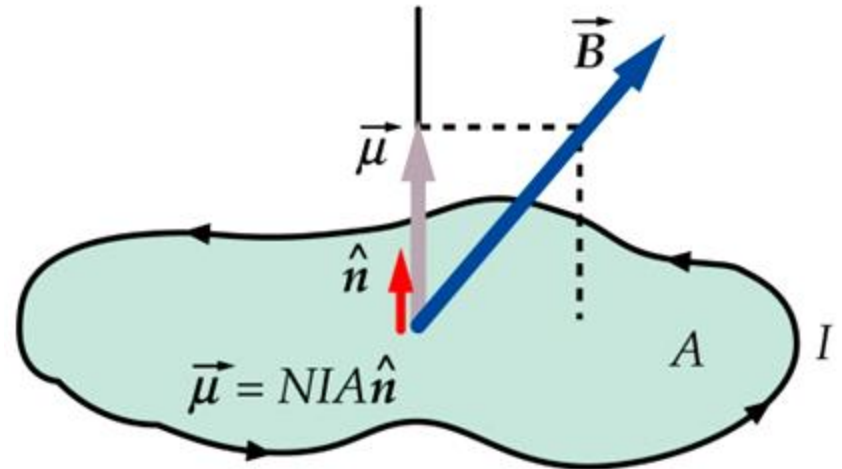
Magnetic moment



Under the influence of a torque, the frame takes a position perpendicular to the direction of vector $\vec{B}$ of the magnetic field so as

$$\vec{\mu} \parallel \vec{B}$$

Magnetic moment is defined for any closed circuit carrying a current I , as:



$$\vec{\mu} = NIA\hat{n}$$

Number of turns

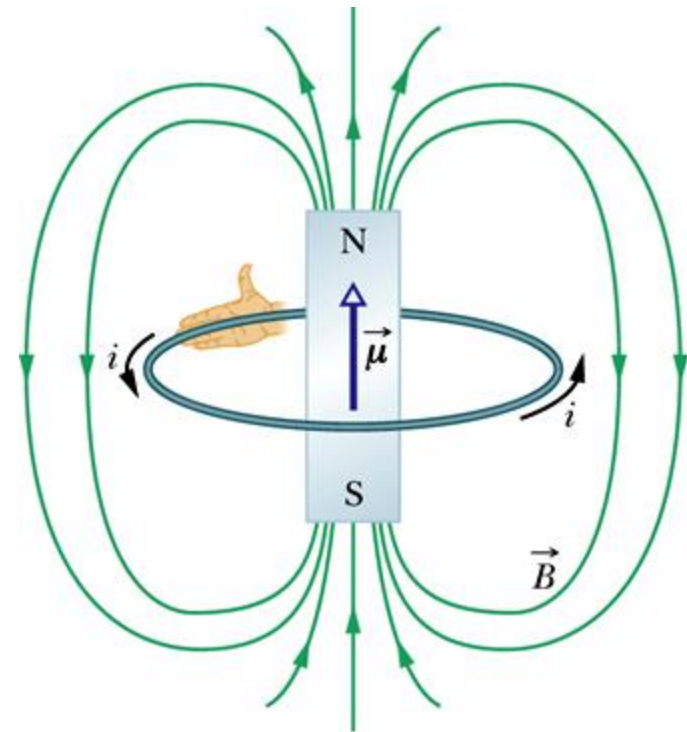
area

Unit vector perpendicular (normal) to the surface A

Magnetic dipole

Magnetic moment characterizes each magnetic dipole. Dipole can be represented not only as a frame (loop, coil) carrying an electric current. The following examples can be given:

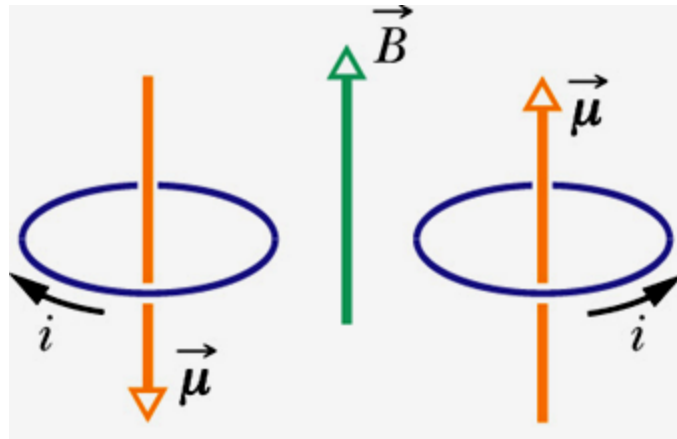
- Permanent magnet ($\mu \approx 5 \text{ J/T}$)
- Earth (approximately) $\mu \approx 8.0 \cdot 10^{22} \text{ J/T}$
- Particles such as an electron ($\mu \approx 9.3 \cdot 10^{-24} \text{ J/T}$),
proton ($\mu \approx 1.4 \cdot 10^{-26} \text{ J/T}$), neutron



Potential energy E_p of magnetic dipole in the external magnetic field is expressed as:

$$E_p = -\vec{\mu} \circ \vec{B}$$

the highest
energy E_p

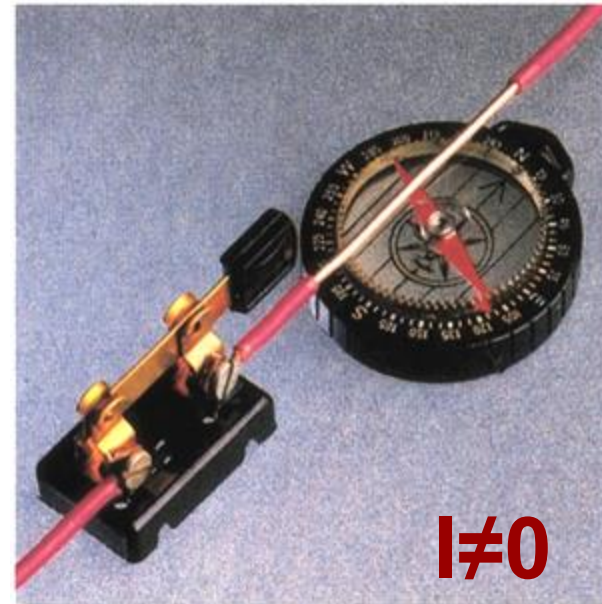
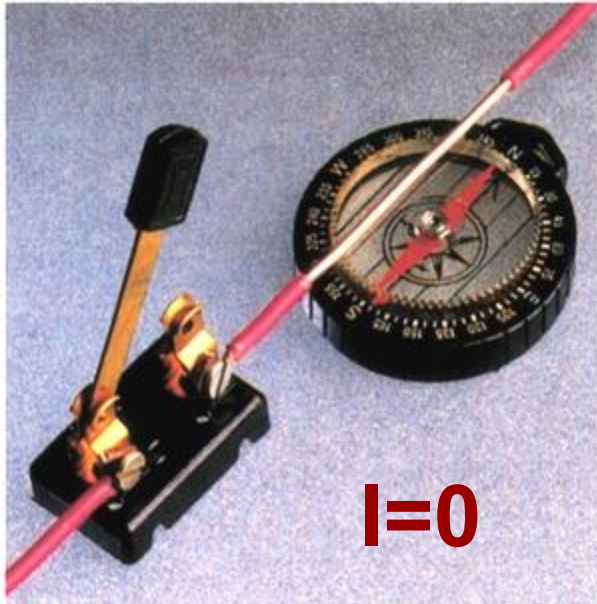


the lowest
energy E_p

Remainder: for an electric dipole in an external electric field

$$E_p = -\vec{p} \circ \vec{E}$$

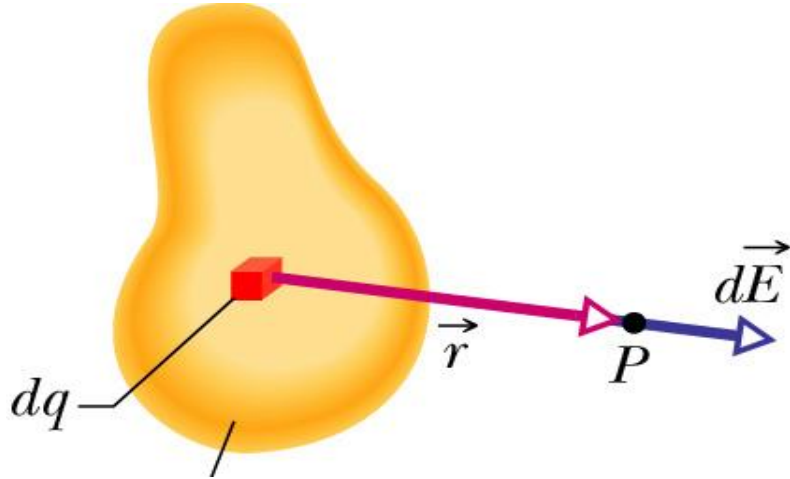
Electric current as a source of magnetic field - Oersted's experiment (1819-1820)



When there is no electric current, a compass needle orients itself parallel to the direction of magnetic field of Earth

When there is electric current, the compass needle is deflected from the direction of magnetic field of Earth. Why?

Law of BIOT and SAVART



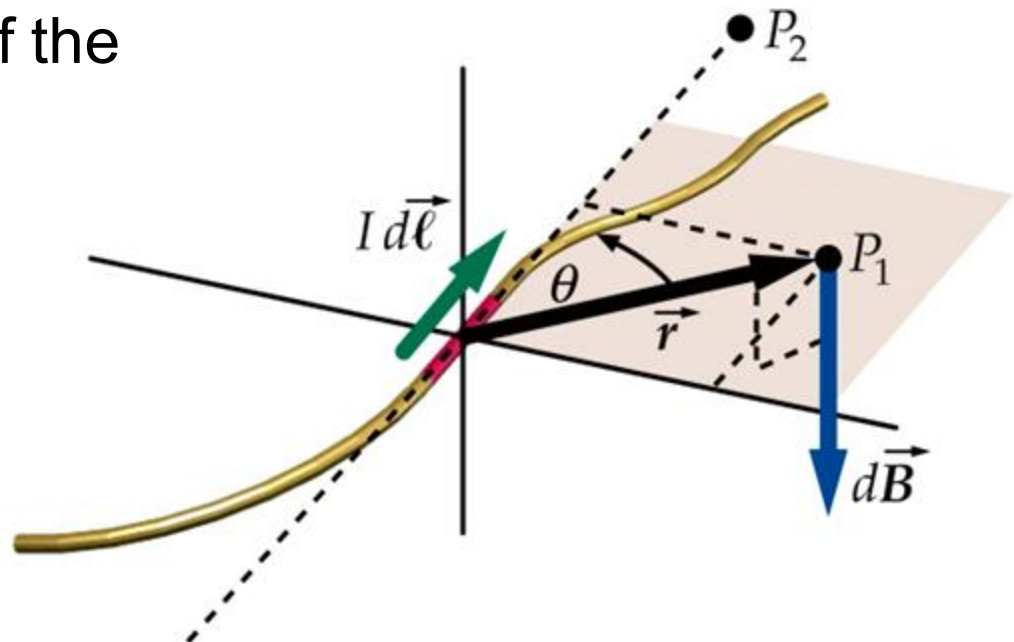
$$d\vec{B} = \frac{\mu_0}{4\pi} \frac{I d\vec{\ell} \times \vec{r}}{r^3}$$

Law of Biot and Savart
Principle of superposition

Continuous distribution of the
electric charge

Coulomb's law

Principle of superposition



Example 10.1

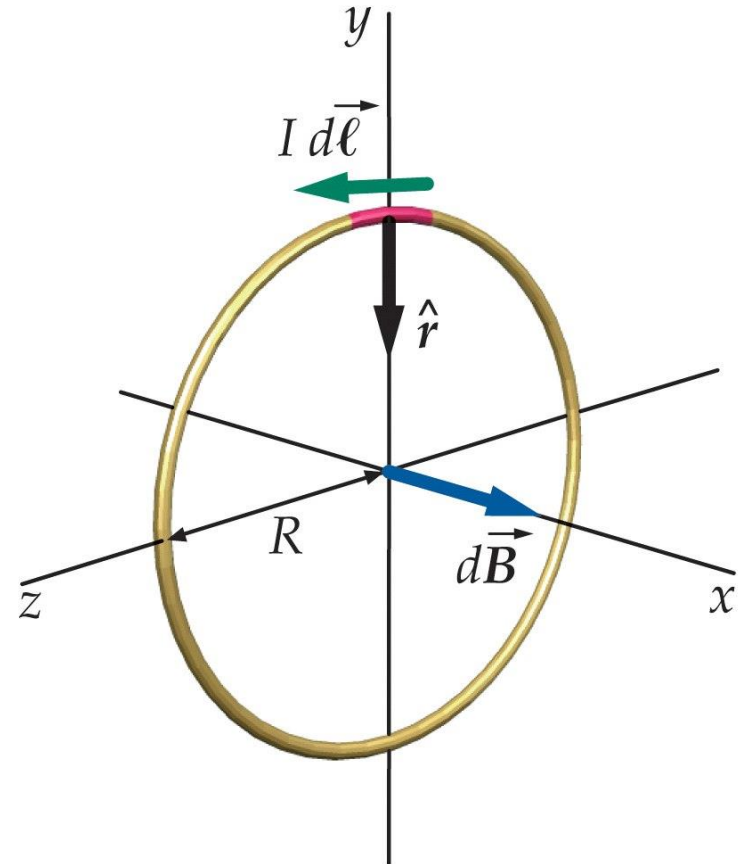
Find the magnetic induction vector exactly at the center of the circular loop of the radius R , carrying the electric current I .

From the law of Biot and Savart

$$dB = \frac{\mu_o}{4\pi} \frac{I dl \sin \theta}{R^2} \quad \theta = 90^\circ$$

$$B = \oint dB = \oint \frac{\mu_o}{4\pi} \frac{I dl}{R^2} = \frac{\mu_o}{4\pi} \frac{I}{R^2} \underbrace{\oint dl}_{2\pi R}$$

$$B = \frac{\mu_o I}{2R}$$



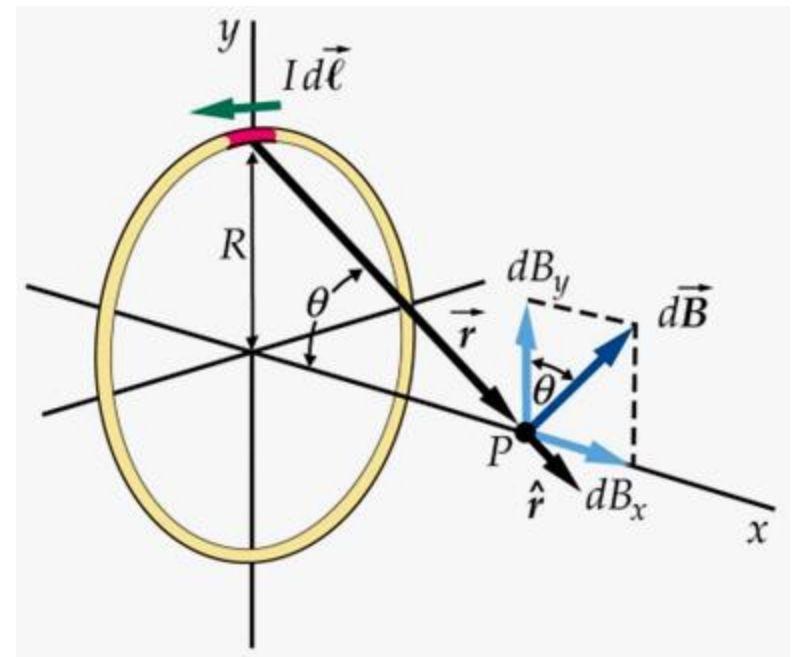
Homework 10.3

Show that a) magnitude of the vector of the magnetic induction at point P lying along the axis of the circular loop carrying the electric current I , is given by

$$B_x = \frac{\mu_o}{2} \frac{IR^2}{(x^2 + R^2)^{3/2}}$$

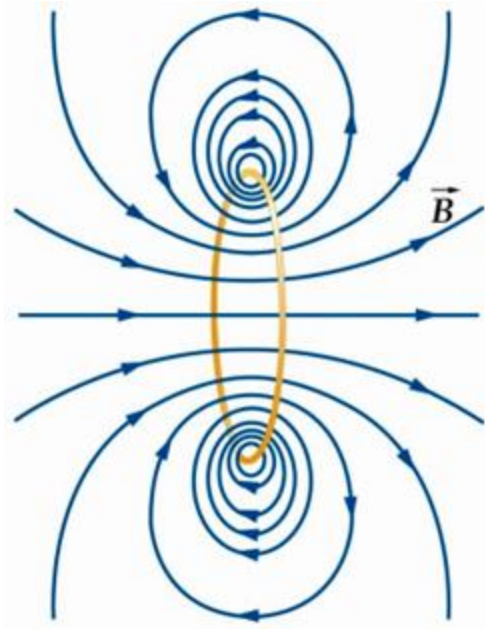
(b) but at a long distance from the center of the loop $x \gg R$

$$B_x = \frac{\mu_o}{2\pi} \frac{\mu}{|x|^3}$$



The loop carrying the electric current behaves as a magnetic dipole – conclusion drawn from 10.3

Circular loop carrying an electric current produces the magnetic field that at large distances resembles that of a magnetic dipole (magnetic field decays with the distance x as x^{-3} , similar dependence can be demonstrated for an electric dipole)



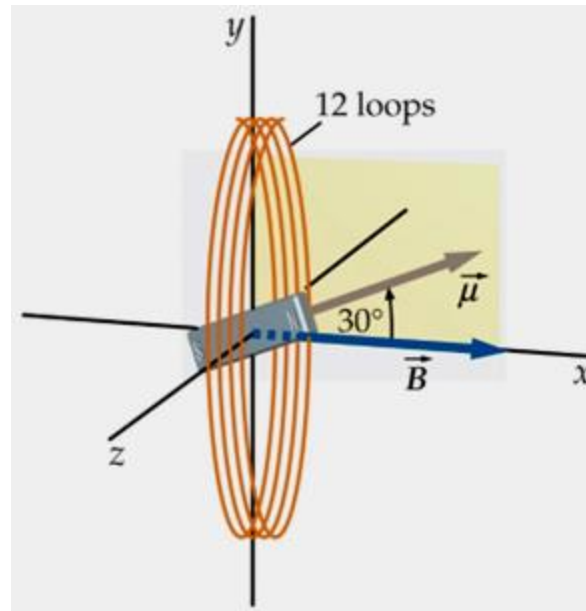
Homework 10.4

A current-carrying circular loop has a radius $r=5$ cm and 12 turns. It carries a current of 4A. The reference frame was chosen in such a way that the loop is entirely enclosed within YZ plane ($x=0$) and its origin is at the center of the loop. Find the magnitude of magnetic induction vector along OX axis at:

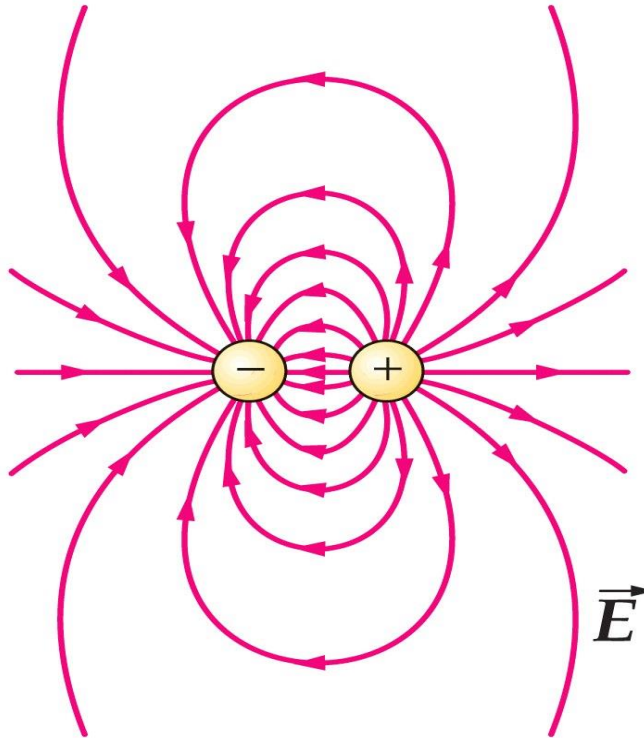
- (a) $x=0$
- (b) $x=3$ cm
- (c) $x=15$ cm
- (d) $x=3$ cm in dipole approximation

Homework 10.5

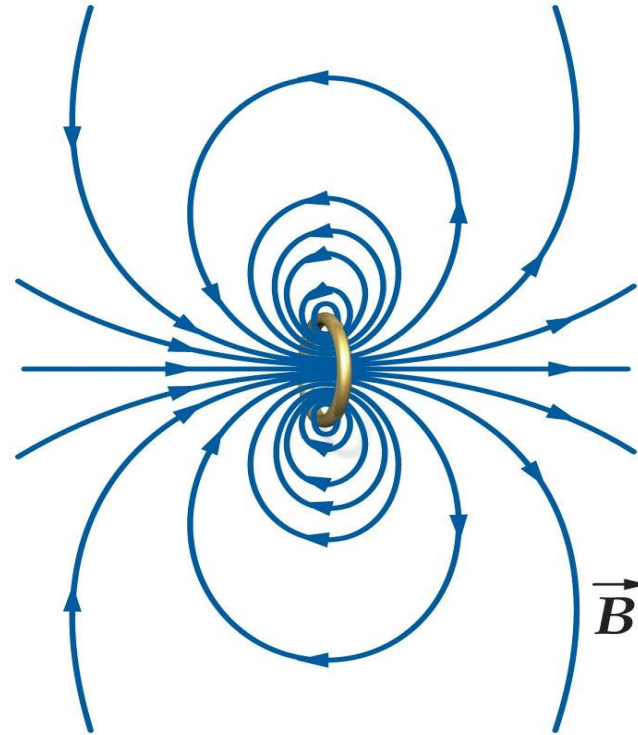
A small bar magnet of a magnetic moment μ is placed exactly at the center of current-carrying loop from the previous problem. Magnetic moment vector is in XY plane and forms an angle of 30° with OX axis. Find a torque acting on the bar magnet. Neglect the effects due to the changes in the magnetic field in the region where the bar magnet is placed.



Magnetic and electric field -similarities and differences

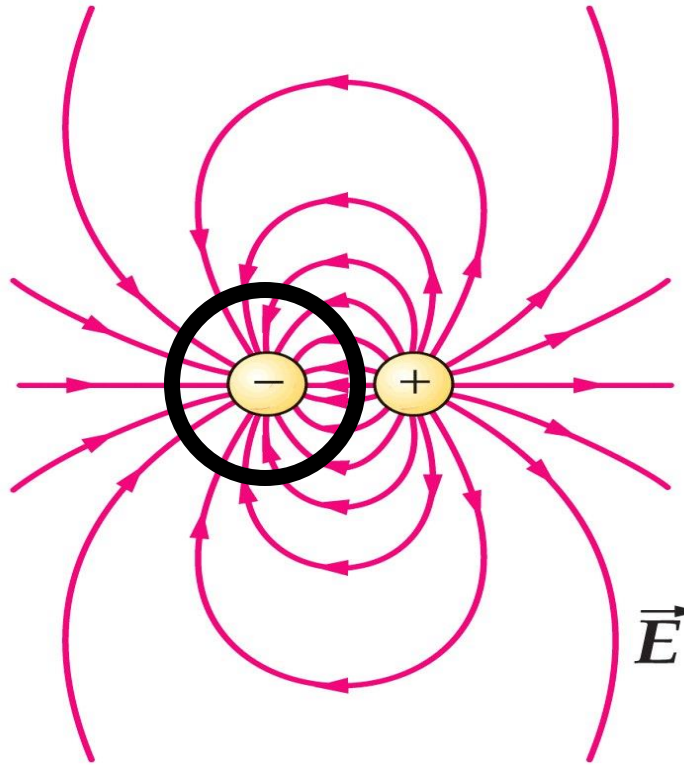


Electric field lines begin and end on the electric charges.



Magnetic field lines form closed loops. They have no beginning and no end.

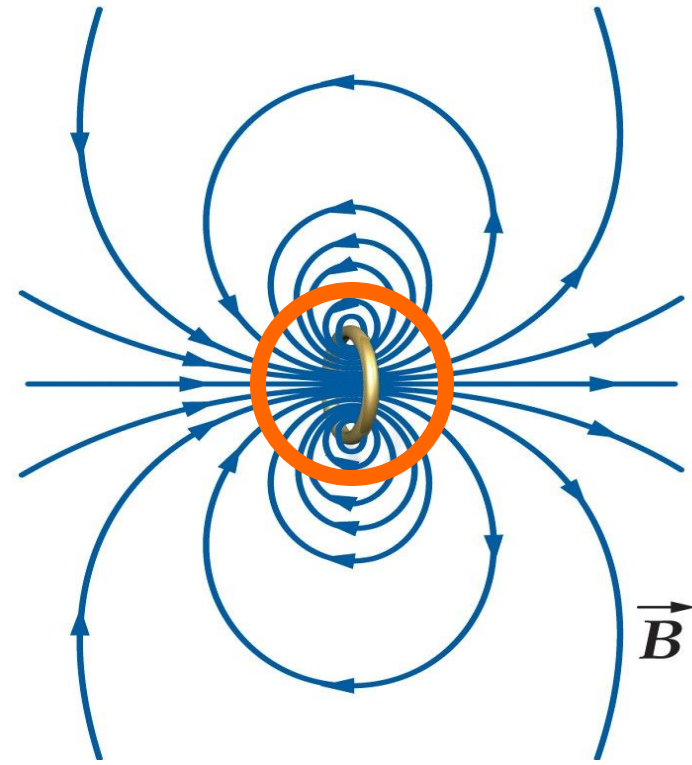
GAUSS' LAW



A single point charge – an electric monopole exists

$$\oint_S \vec{E} \cdot d\vec{A} = \frac{q}{\epsilon_0}$$

$$\text{div } \vec{E} = \frac{\rho}{\epsilon_0}$$



$$\oint_S \vec{B} \cdot d\vec{A} = 0 \quad \text{div } \vec{B} = 0$$

No magnetic monopoles. A magnet or a loop carrying a current are magnetic dipoles

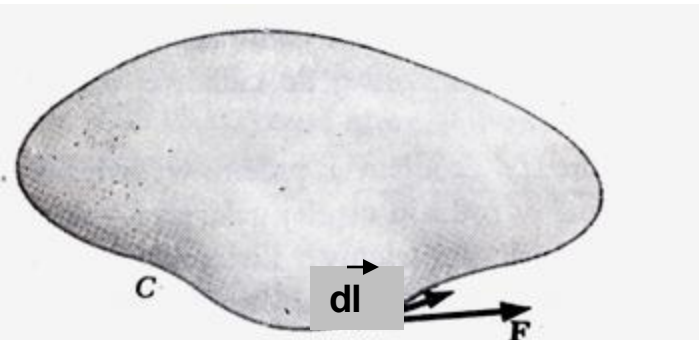
Gauss' law for magnetic field is one of **Maxwell** equations.

It states that magnetic field is **divergence-free (incompressible)**.
The flux of magnetic field through any closed surface is zero.
One cannot extract a single magnetic pole – **magnetic monopoles do not exist**.

$$\Phi_B = \oint_S \vec{\mathbf{B}} \cdot d\vec{\mathbf{A}} = 0$$

$$\text{div } \vec{\mathbf{B}} = 0$$

CIRCULATION OF VECTOR FIELD



Circulation of vector field $\vec{F}$ along a closed loop is defined as a line integral:

$$\Gamma = \oint_C \vec{F} \circ d\vec{l}$$

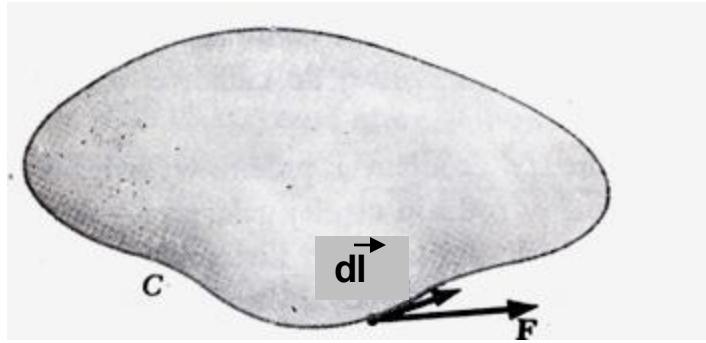
$d\vec{l}$ the element of integration path has a direction of tangent to the curve C at a given point

If $\vec{F}$ represents a force, then circulation Γ has a physical sense of work.

If $\vec{F}$ is a conservative force (due to electrostatic or gravitational field), then $\Gamma=0$.

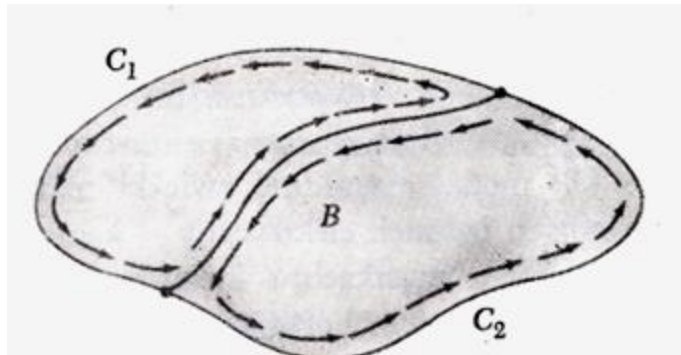
Curve C encloses a certain surface, bounded by this curve.

CURL OF THE FIELD



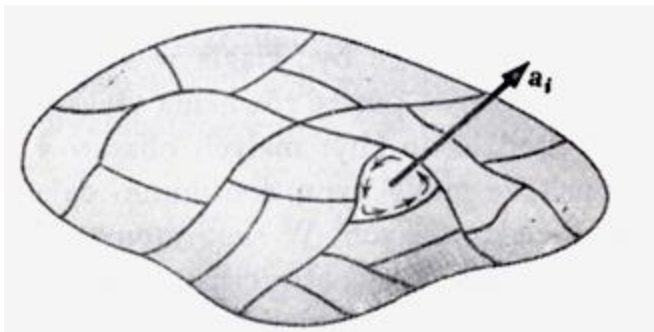
$$\Gamma = \oint_C \vec{F} \circ d\vec{I}$$

Tracing a curve B, we create two closed loops C_1 and C_2 so as:



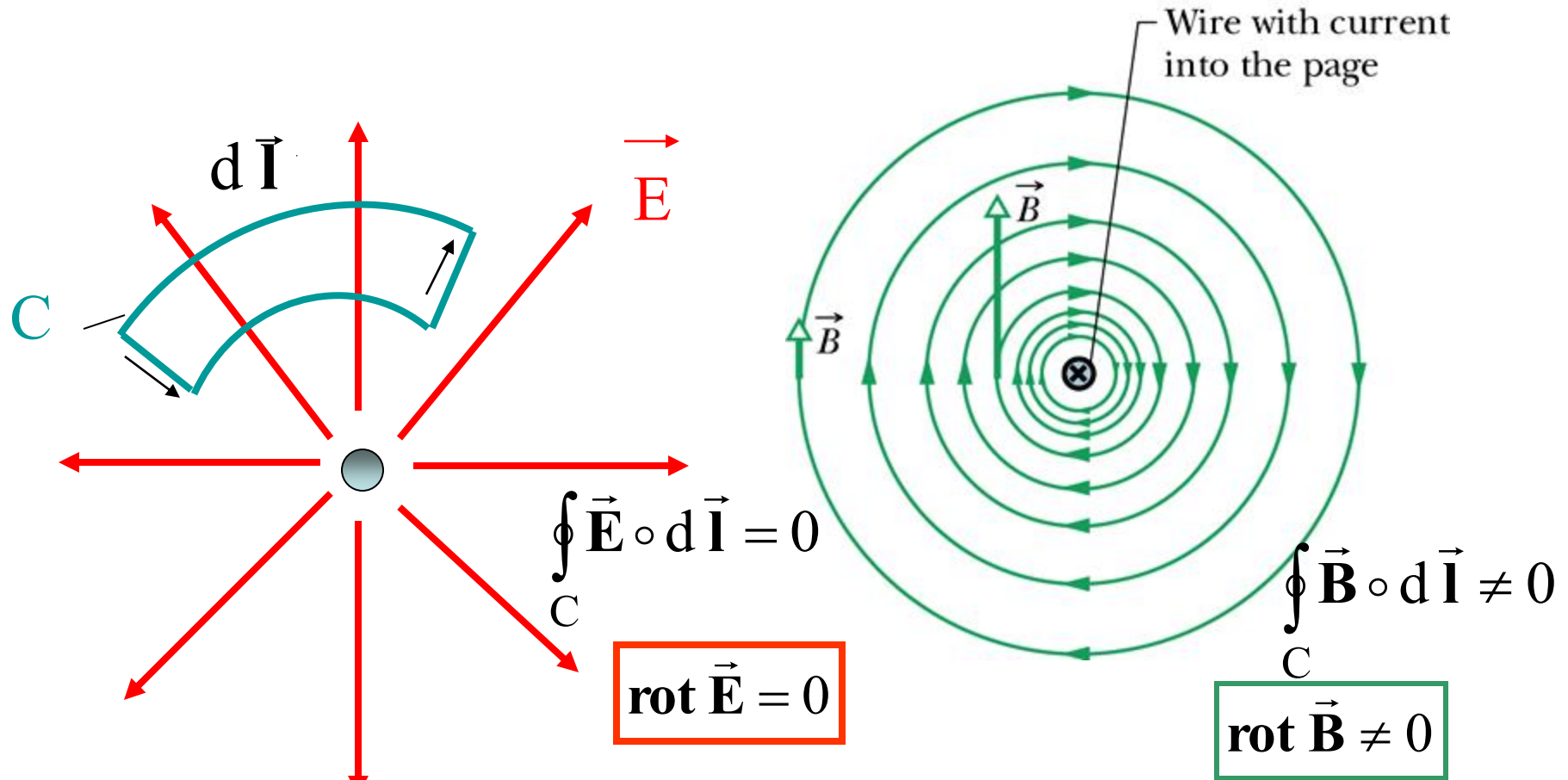
$$\oint_C \vec{F} \circ d\vec{I} = \oint_{C_1} \vec{F} \circ d\vec{I} + \oint_{C_2} \vec{F} \circ d\vec{I}$$

Definition of the curl (rotation) operator



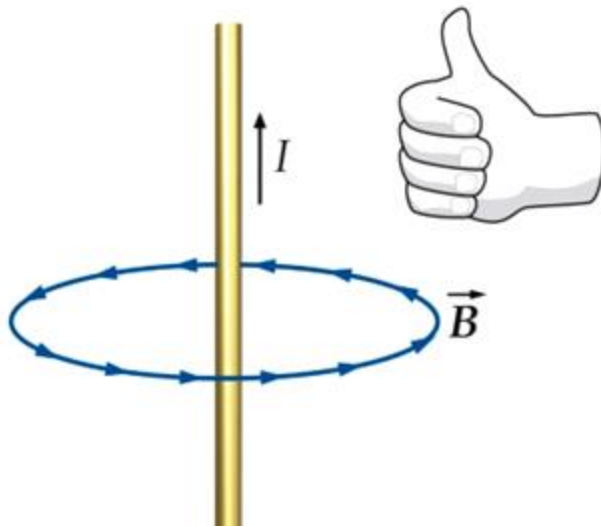
$$(\text{rot } \vec{F}) \circ \hat{n} = \lim_{a_i \rightarrow 0} \frac{\oint_{C_i} \vec{F} \circ d\vec{I}}{a_i}$$

Question: Electrostatic field is **irrotational** (field rotation is zero at each point). What about a rotation of a magnetic field?



Answer: In fact, magnetic field is not irrotational. Ampère law?

Ampère law



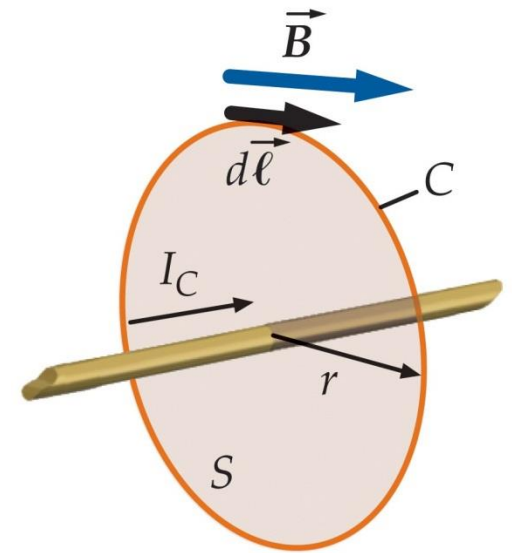
$$\oint_C \vec{B} \circ d\vec{l} = \mu_0 I_C$$

circulation of
magnetic field

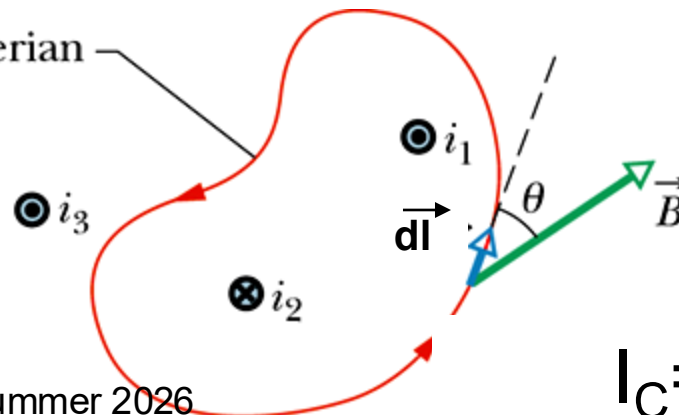
current inside
integration path C

μ_0 – magnetic permeability of
vacuum, universal constant

$$\mu_0 = 4\pi \cdot 10^{-7} \text{ T} \cdot \text{m} / \text{A}$$



Amperian
loop



$$I_C = i_1 - i_2$$

Stokes theorem

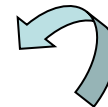
- Relates circulation of a vector around a curve C to curl at a given point, in a similar way as Gauss-Ostrogradsky theorem relates a flux emanating from a surface to a divergence at a point

$$\oint_C \vec{F} \circ d\vec{l} = \iint_S (\text{rot } \vec{F}) \circ d\vec{a}$$

Surface integral, S is a surface bounded by the curve C

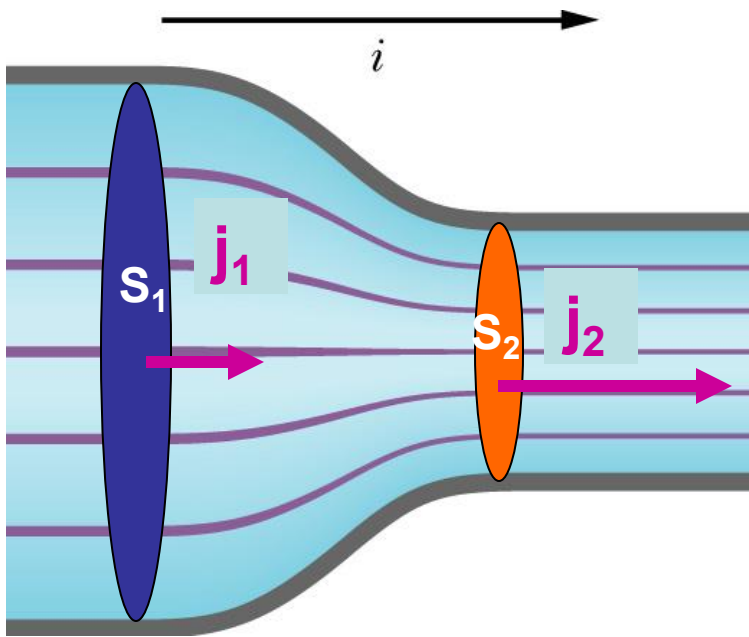
- Differential form of Ampère law

$$\text{rot } \vec{B} = \mu_0 \vec{j}$$



Current density

- Current density is a vector of magnitude equal to the electric current crossing the elemental cross-section of the conductor per its infinitesimal area dS



$$\mathbf{j} = \frac{d\mathbf{i}}{dS}$$

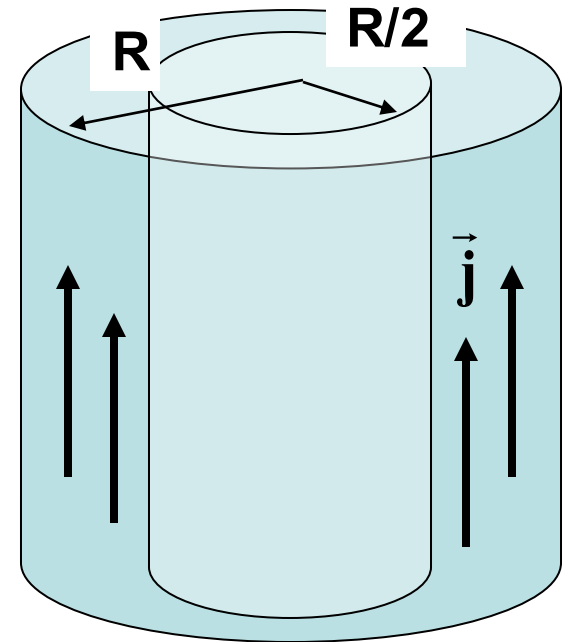
Flux of current density

$$i = \int \overbrace{\vec{j} \circ d\vec{S}}$$

Magnitude of the electric current remains constant, current density changes – analogous to the conservation of mass, conservation of charge (no accumulation, no leak)

Example 10.2

Current density in the cylinder of the radius $R = 2 \text{ mm}$ is the same at each point of the cross section of the conductor and equal to $j = 2 \cdot 10^5 \text{ A/m}^2$. Calculate the current in the external shell of the cylindrical conductor, i.e., in the region between $R/2$ and R .



Solution:

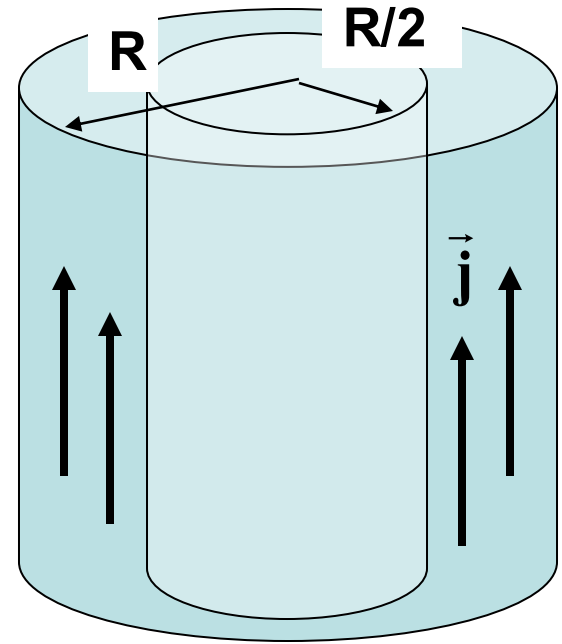
$i = jS'$ as $j = \text{const}$ on the cross-sectional area

$$S' = \pi R^2 - \pi \left(\frac{R}{2} \right)^2 = \frac{3\pi}{4} R^2$$

Finally: $i = 1.9 \text{ A}$

Example 10.3

Assume that the current density in a cylindrical conductor of the radius $R = 2$ mm is not constant but changes with the distance r from its center according to $\vec{j} = \alpha r^2$, where $\alpha = 3 \cdot 10^{11} \text{ A/m}^4$. Calculate the current in the external shell of the cylindrical conductor, i.e., in the region between $R/2$ and R .



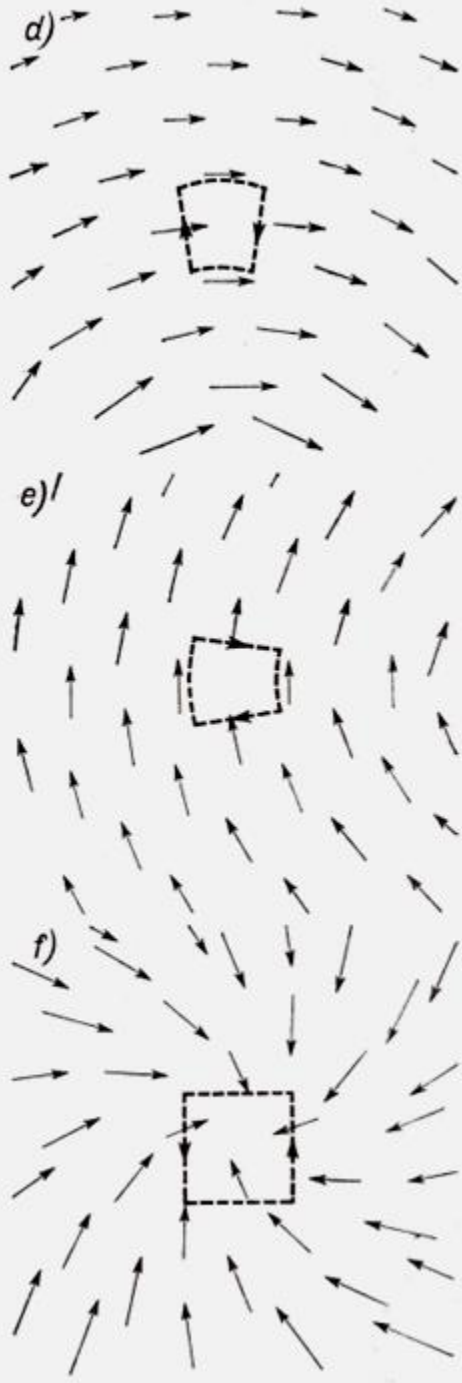
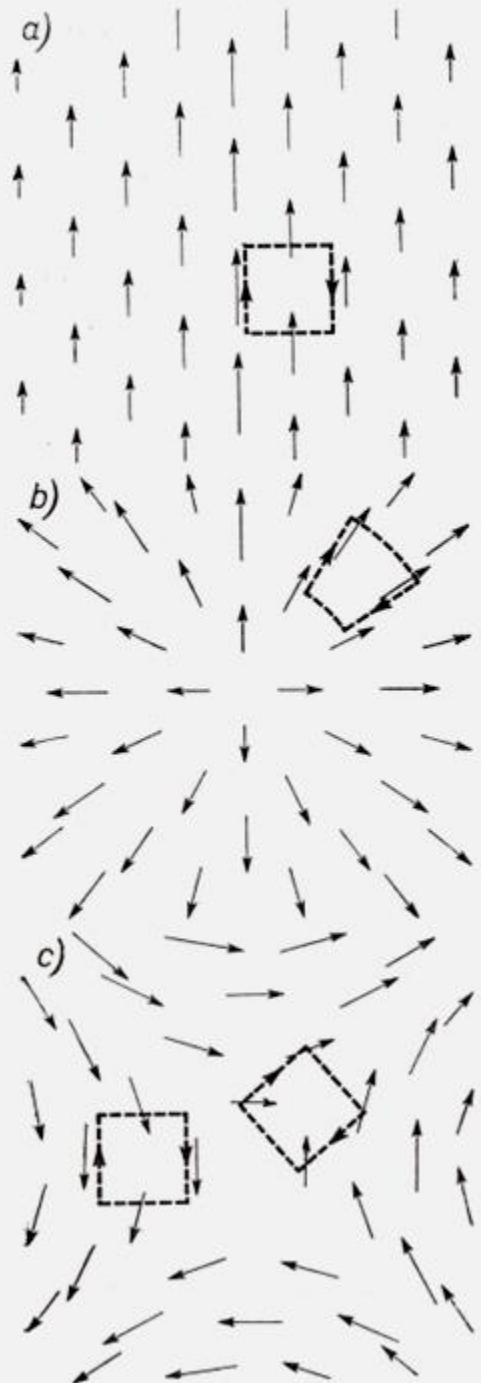
Solution:

$$i = \int \vec{j} \cdot d\vec{S} = \int j dS = \int_{R/2}^R \alpha r^2 2\pi r dr$$

Finally:

$$i = \frac{15}{32} \pi \alpha R^4 \quad i = 7.1 \text{ A}$$

Homework 10.6

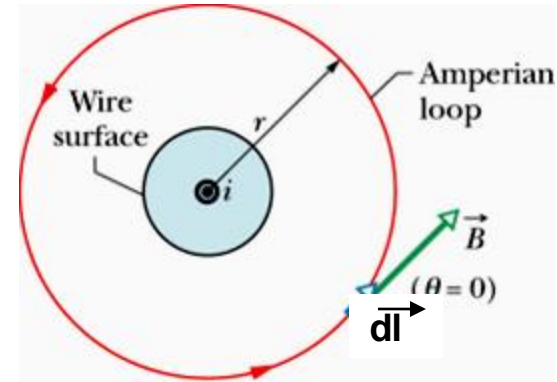


Four of the vector fields shown here are incompressible, while three of them are irrotational. Can you find them?

Applications of Ampère's Law

- Magnetic field around a long, straight wire

$\vec{B} \parallel d\vec{l}$ $B = \text{const}$ along a curve C
(integration path = a circle of radius r)



Circulation of the magnetic field vector around the wire

$$\oint_C \vec{B} \circ d\vec{l} = B \oint_C dl = 2\pi r B$$

Using Ampère's law

$$2\pi r B = \mu_0 i$$

Finally:

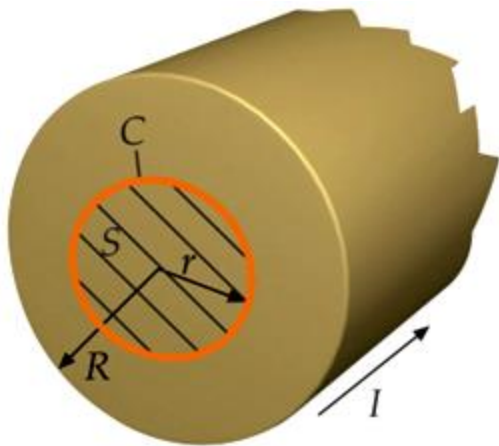
$$B = \frac{\mu_0 i}{2\pi r}$$

Homework 10.7

The same result can be obtained by a tiresome integration, using the law of Biot and Savart. Try this approach in order to appreciate later the Ampère's law. The solution to this problem can be found in many textbooks such as Halliday, Resnick, Fundamental of Physics.

Applications of Ampère's Law

- Magnetic field inside the long, straight wire of circular cross-section of a radius R , carrying a current I ($r < R$)



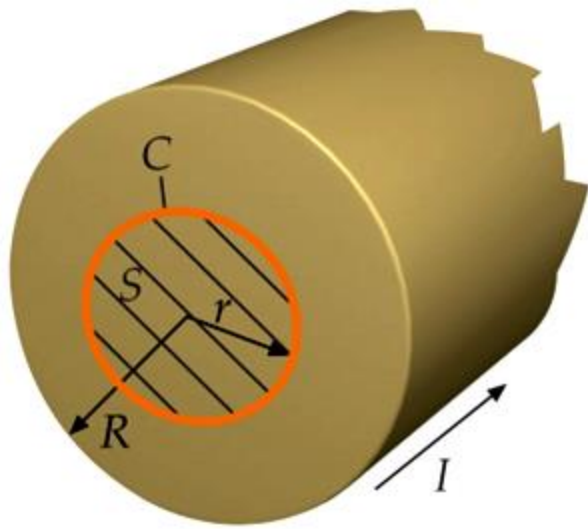
Circulation of the magnetic field vector along a circle of radius r can be expressed in both cases $r < R$ i $r > R$ as

$$\oint_C \vec{B} \circ d\vec{l} = 2\pi r B$$

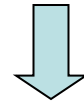
In this case one has to find the current I_C inside the Amperian loop

Current density j is constant

$$j = \frac{I_C}{\pi r^2} = \frac{I}{\pi R^2}$$



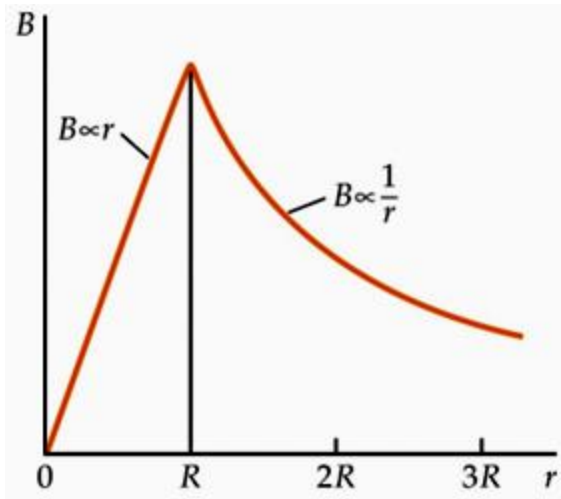
$$j = \frac{I_C}{\pi r^2} = \frac{I}{\pi R^2}$$



$$I_C = \frac{I}{R^2} r^2$$

From the Ampère's law

$$\oint_C \vec{B} \cdot d\vec{l} = \mu_0 I_C$$



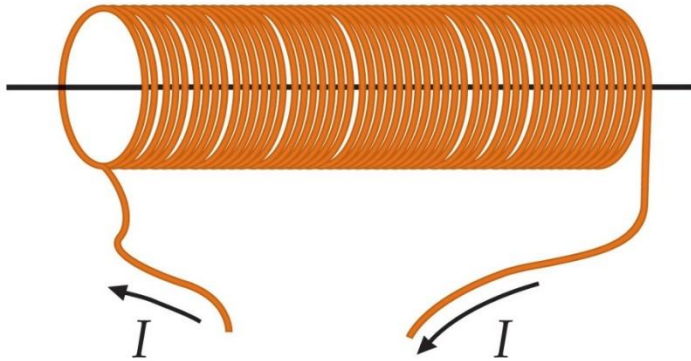
$$2\pi Br = \mu_0 \frac{I}{R^2} r^2$$

Answer:

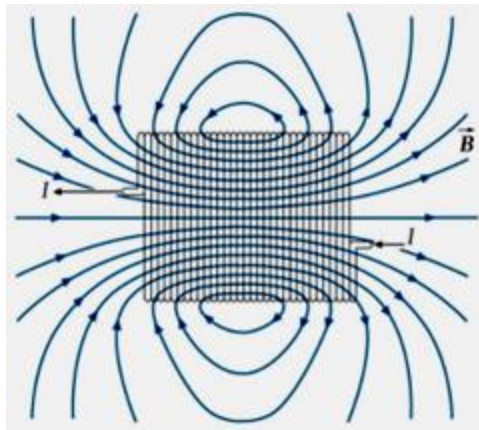
$$B = \frac{\mu_0 I}{2\pi R^2} r$$

Applications of Ampère's Law

- Magnetic field inside a solenoid



Solenoid produces a **uniform** magnetic field and plays an analogous role as a plane-parallel capacitor in electrostatics.



Lecture 10

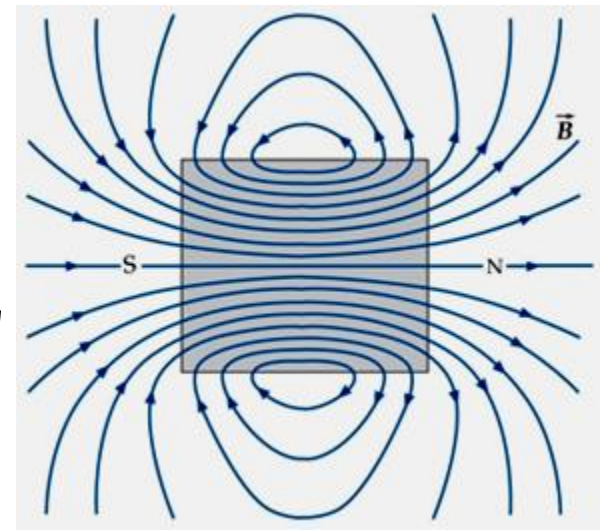
solenoid



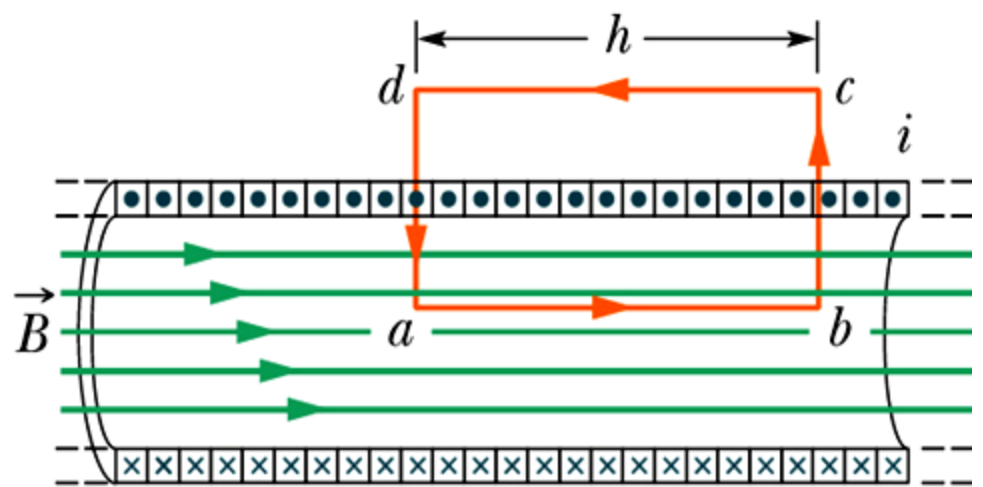
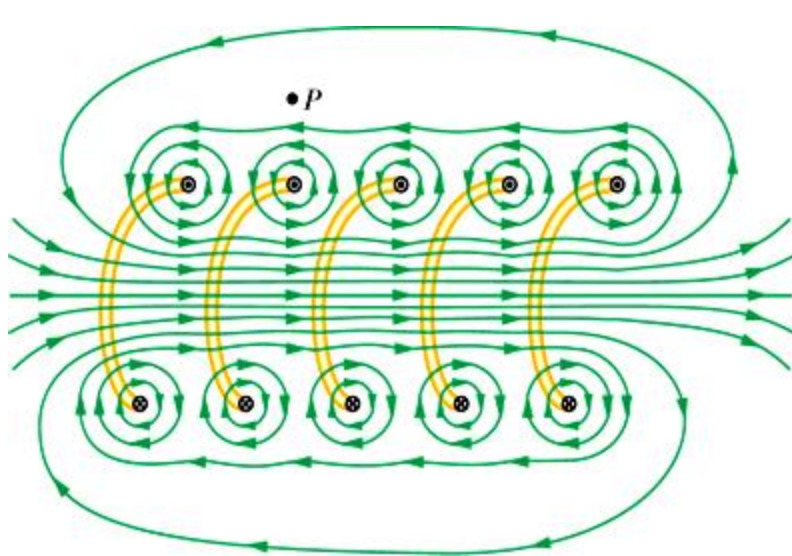
bar magnet



summer 2026



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$$\oint_C \vec{B} \circ d\vec{l} = \underbrace{\int_a^b \vec{B} \circ d\vec{l}}_{Bh} + \underbrace{\int_b^c \vec{B} \circ d\vec{l}}_0 + \underbrace{\int_c^d \vec{B} \circ d\vec{l}}_0 + \underbrace{\int_d^a \vec{B} \circ d\vec{l}}_0$$

why?

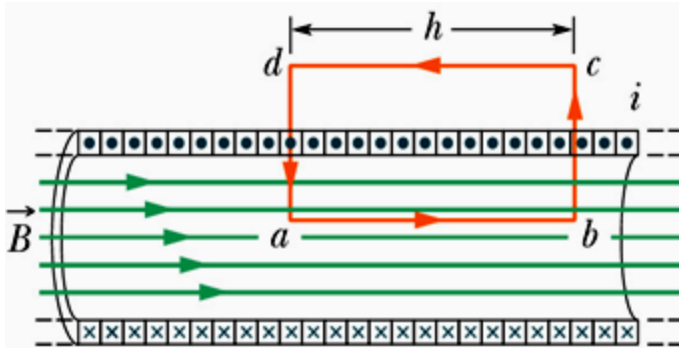
$$\vec{B} \parallel d\vec{l}$$

$$\vec{B} \perp d\vec{l}$$

$$B=0$$

$$\vec{B} \perp d\vec{l}$$

uniform field



$$\oint_C \vec{B} \circ d\vec{l} = Bh = \mu_o I_C$$

$$I_C = (nh)i$$

turn density of the solenoid,
i.e., the number of turns per
unit length

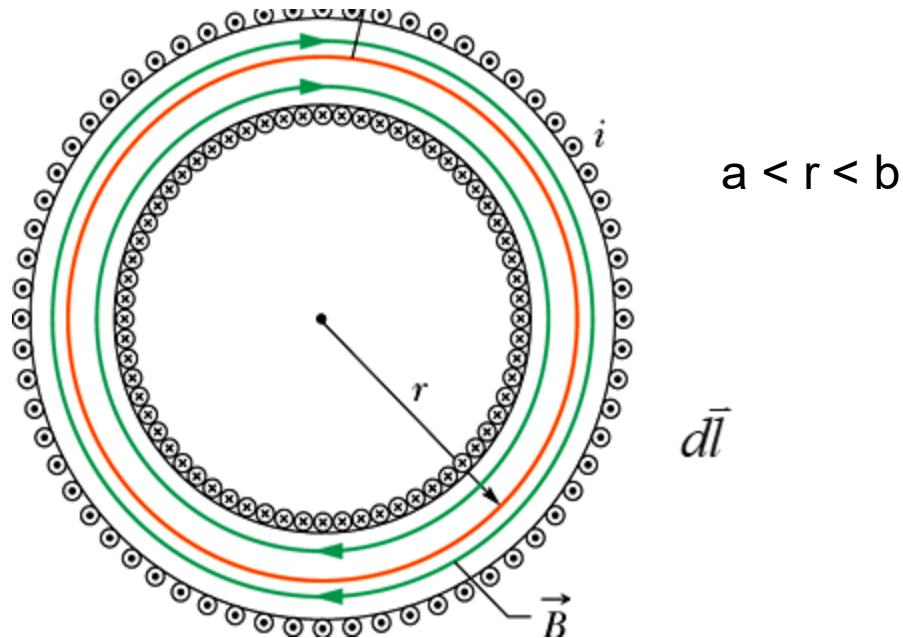
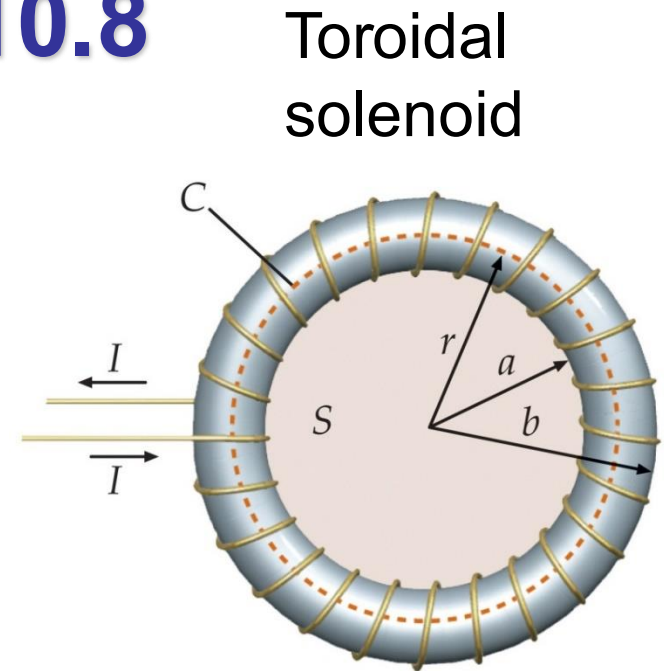
current in the solenoid

ideal solenoid

$$B = \mu_o ni$$

Homework 10.8

Apply the Ampère's law to find the magnitude of the magnetic vector inside the toroidal solenoid carrying a current I



$$B = \frac{\mu_0 NI}{2\pi r}$$

N – number of turns

Summary

- Motion of electric charge in magnetic and electric fields takes place under the Lorentz force. Magnetic field does not change the kinetic energy of a charged particle (does not accelerate it) but deflects its trajectory.
- Differences between electrostatic and magnetic fields can be demonstrated using Gauss' laws for electricity and magnetism. Magnetic dipoles do not exist.
- Electrodynamic force acts on the conductor carrying an electric current placed in the magnetic field.
- Electric charge at rest produces the electrostatic field while the moving charge (electric current) is a source of the magnetic field.
- Magnitude of the magnetic field vector produced by the carrying current conductor can be calculated from the laws of Biot and Savart or Ampère, but practical application of Ampère's law requires certain symmetry.